

Simulation Study on Performance of **Bayesian Dynamic Borrowing** on DSI Gastric Cancer Clinical Data.

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Project inspired by a 2023 paper: “Bayesian Dynamic Borrowing Of Historical Information With Applications To The Analysis Of Large-Scale Assessments”

Why this project?

DG-01 Study

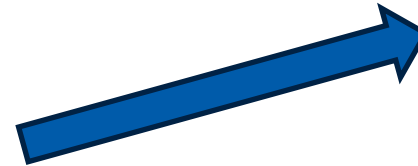
(Phase-2 randomized trial)

- Enrolls 125 patients in T-DXd arm

DG-02 Study

(Phase-2 single arm trial)

- Enrolls 79 patients in T-DXd arm



PROSPERITY

(Gastric Non-interventional Study)

- Plans to enroll 162 patients in T-DXd arm

Same compound (T-DXd) and same outcome of interest studied in all 3 studies:

Aim:

- Increase effective sample size.
- Improve prediction accuracy.
- Reduce estimation bias.

Key Question: To what extent should historical data be used?

Answer: It depends on the similarity between historical and current data.

Using a flexible informed prior distribution for PROSPERITY, based on DG-01 and DG-02 study data, would be a wise attempt.

What is the research question?

Evaluating impact of,

- Sample size
 - 100, 162.
- Degree of differences in two populations.
 - 4 conditions ranging from most homogeneous to most heterogeneous.
- Choice of prior
 - $\beta_1, \beta_2, \beta_3 \sim N(\mu, \text{diagonal}(\tau^2))$
assuming an exchangeability assumption across 3 studies.
$$\mu \sim N(0, 10^2)$$
$$\tau^2 \sim IG(0.1, 0.1),$$
$$IG(1, 1),$$
$$IG(1, 0.1)$$

on the parameter estimation in terms of following measures (all or some of them):

1. Leave-One-Out Cross-Validation Information Criterion.
2. Total Effective Sample Size.
3. Mean Squared Error.
4. Percent Bias.
5. Width of 95% HPD credible intervals.
6. Coverage of 95% HPD credible intervals.

How are we planning to answer it?

- Simulation of the PROSPERITY Data.
 - $Z'_i s \sim MVN(\mathbf{0}, \Sigma)$, $U \sim N(0,1)$
 - Baseline Weibull hazard and Censoring time $\sim \text{Exponential}(\lambda)$.
- Single arm (T-DXd) analysis across 3 studies and regressing the survival outcomes against 6 covariates like
 - Age,
 - Gender,
 - BMI,
 - Liver metastasis,
 - Hepatic impairment,
 - Number of metastatic sites.

- Endpoint of interest: Time to composite outcome i.e., next systemic treatment/death.
- Outcome Measure: Median survival time.
- The results of the simulation will also be compared with the static borrowing methods like
 - Power prior (static version with $\alpha = 0, 0.5, 1$).
 - Complete pooling.
 - Non-Informative (weak priors).

TESS and LOOIC

TESS

$$\begin{aligned} & \text{(Total Patients In Analysis)} \\ & * \left(\frac{\text{Posterior Variance Under Complete Pooling}}{\text{Posterior Variance Under Given Prior Setting}} \right) \end{aligned}$$

LOOIC

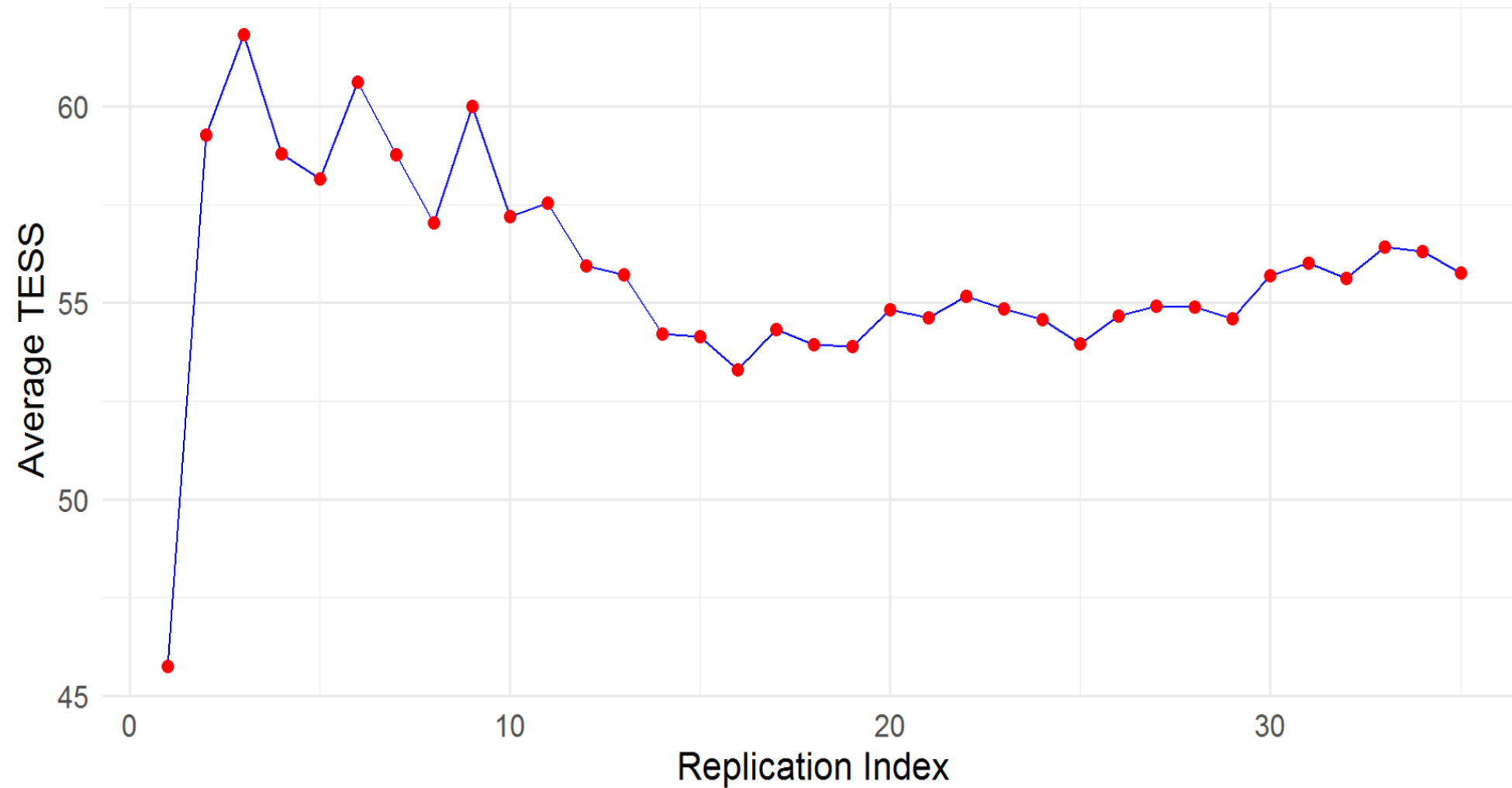
Function of estimated 'Expected Log Point-Wise Predictive Density (ELPD)' for each observation, one at a time.

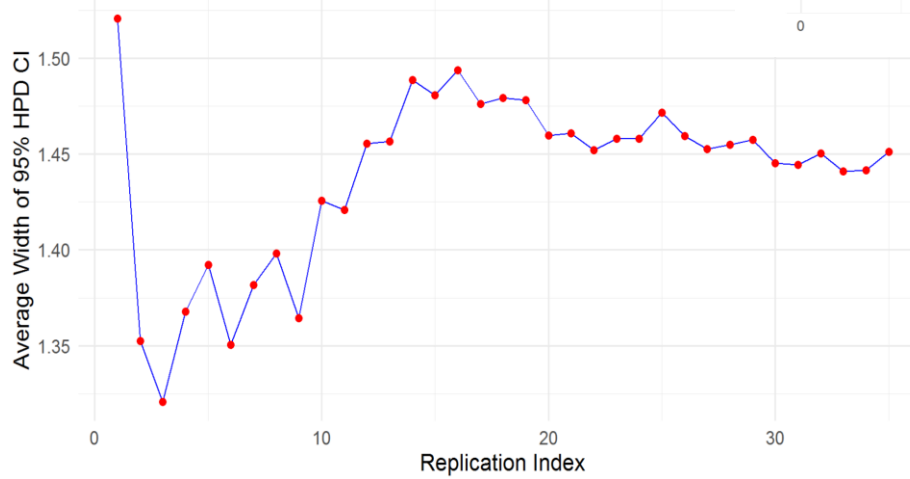
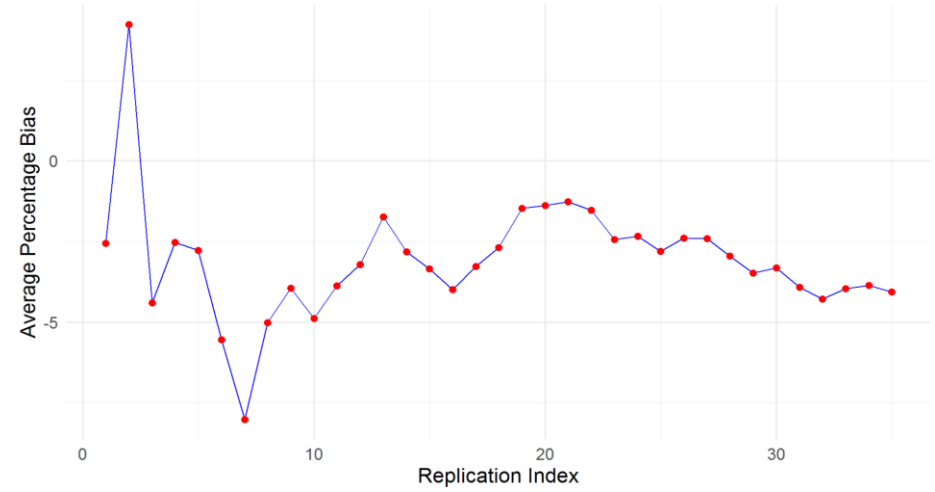
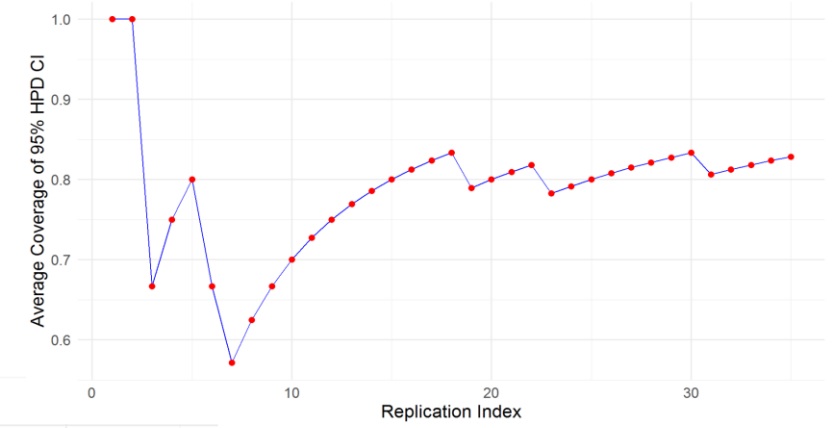
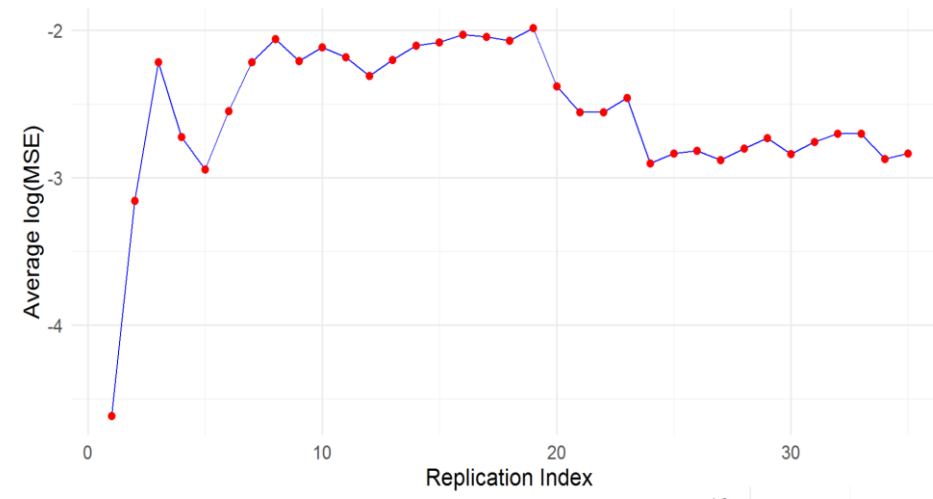
Explain other performance metrics

Details about simulation.

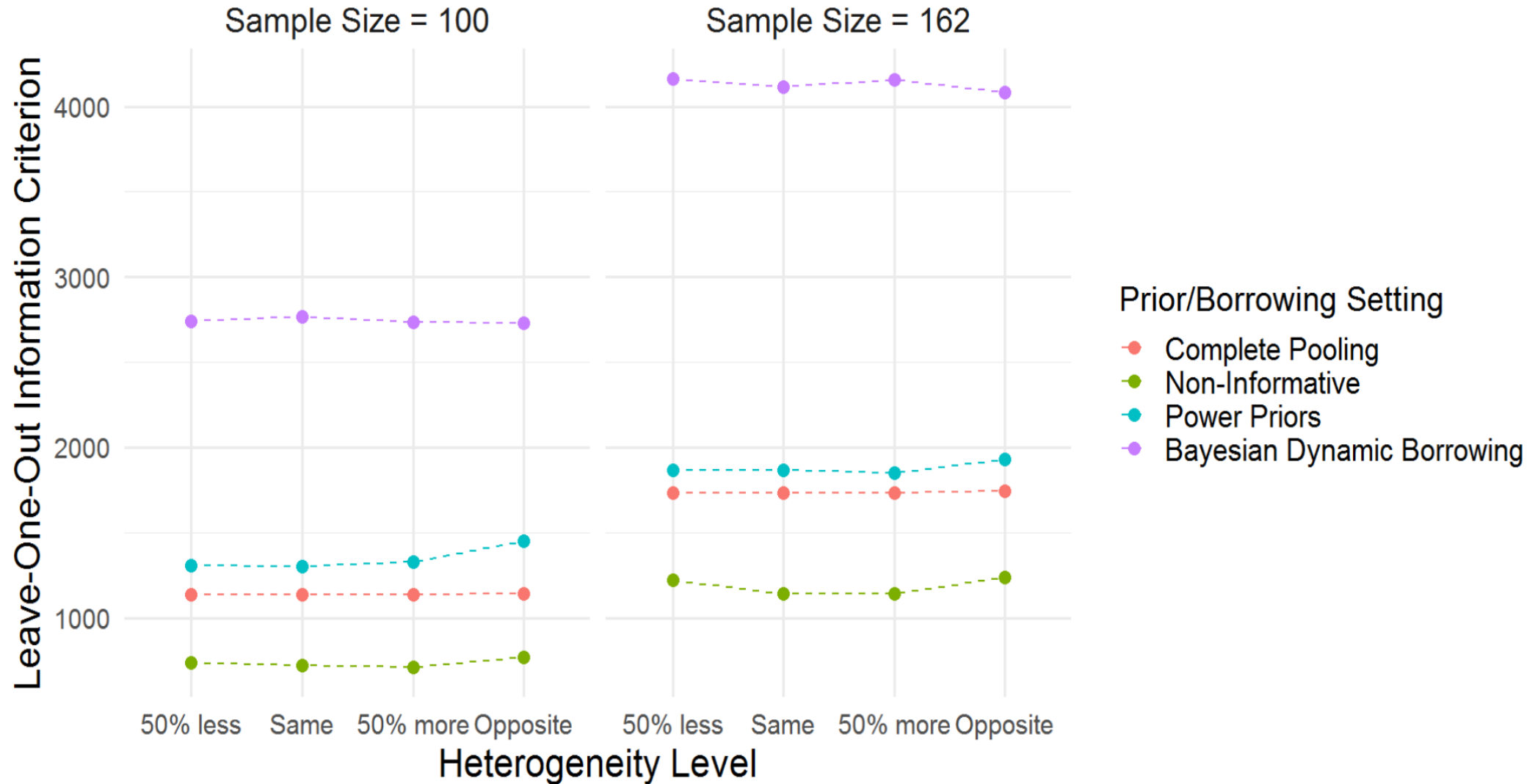
- 10,000 iterations in 3 MCMC chains.
- Number of replications per each combination of Sample Size, Prior Setting, Heterogeneity Condition, and Baseline Covariates (including Intercept) is 35.
- We are interested in the overall performance of the priors; hence effective number of replication is $35 \times 7 = 245$.
- For performance results, we won't be focusing on differentiating between priors within each prior/borrowing setting.

Why only 35 replications?

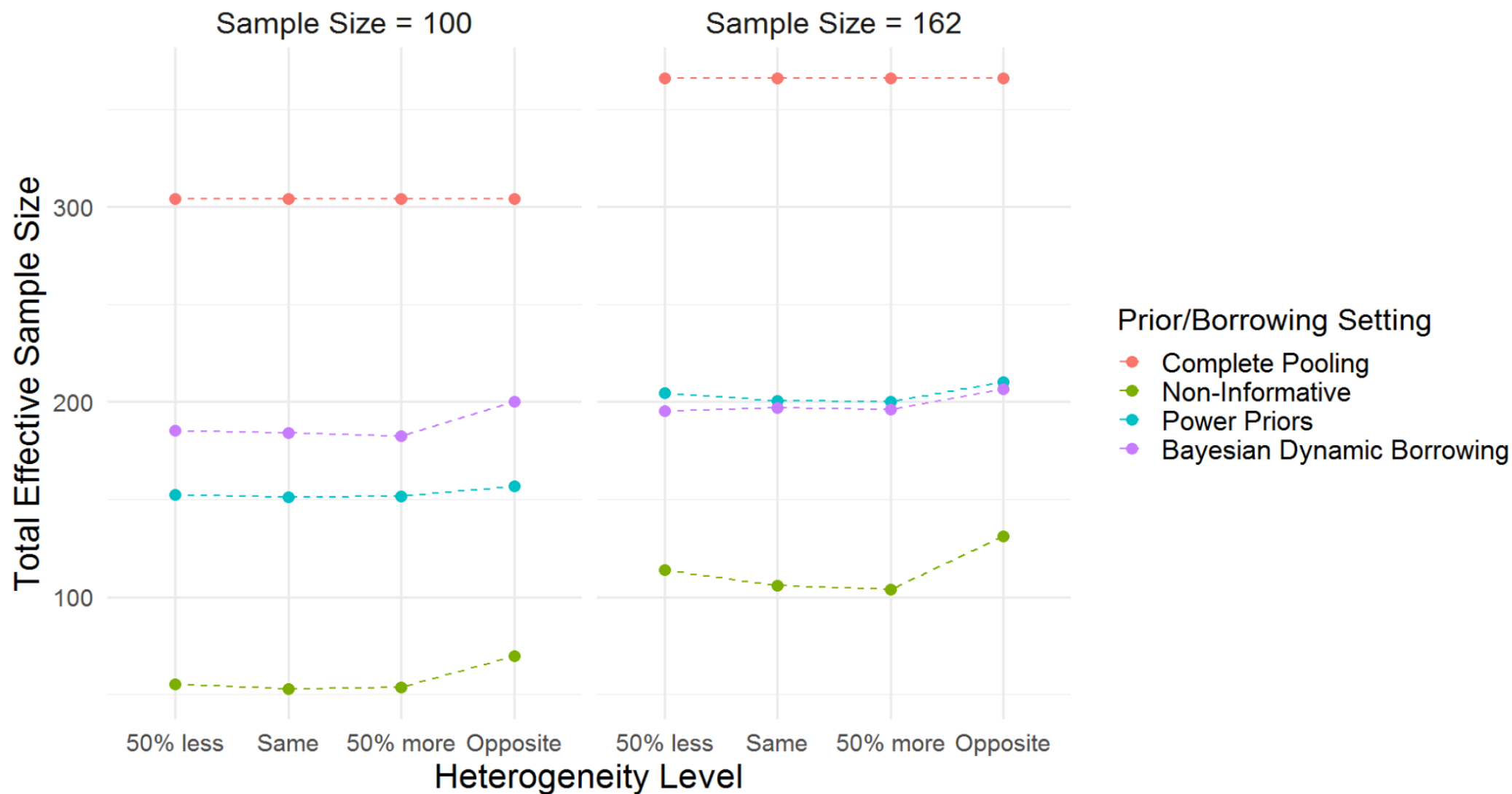




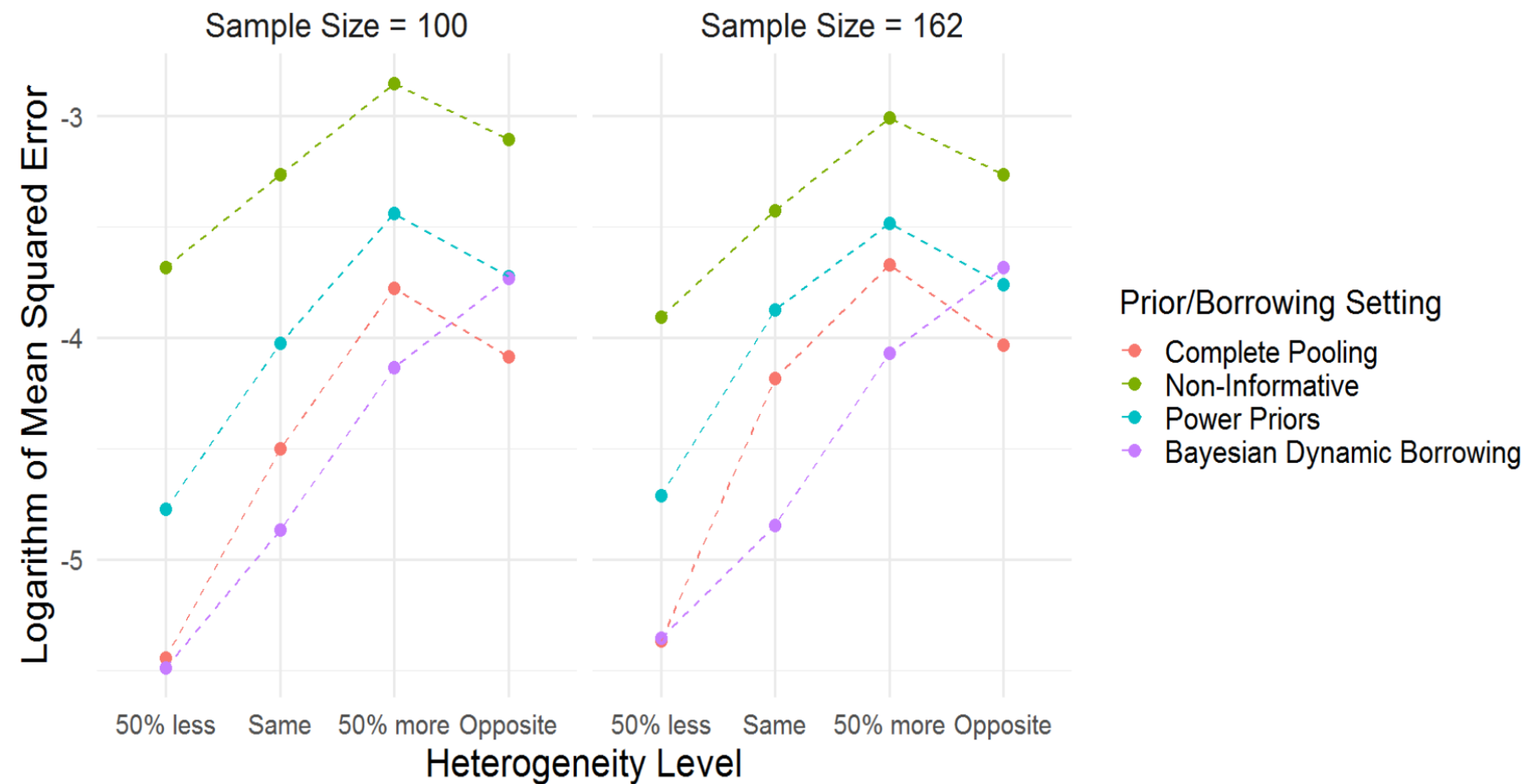
Results - LOOIC



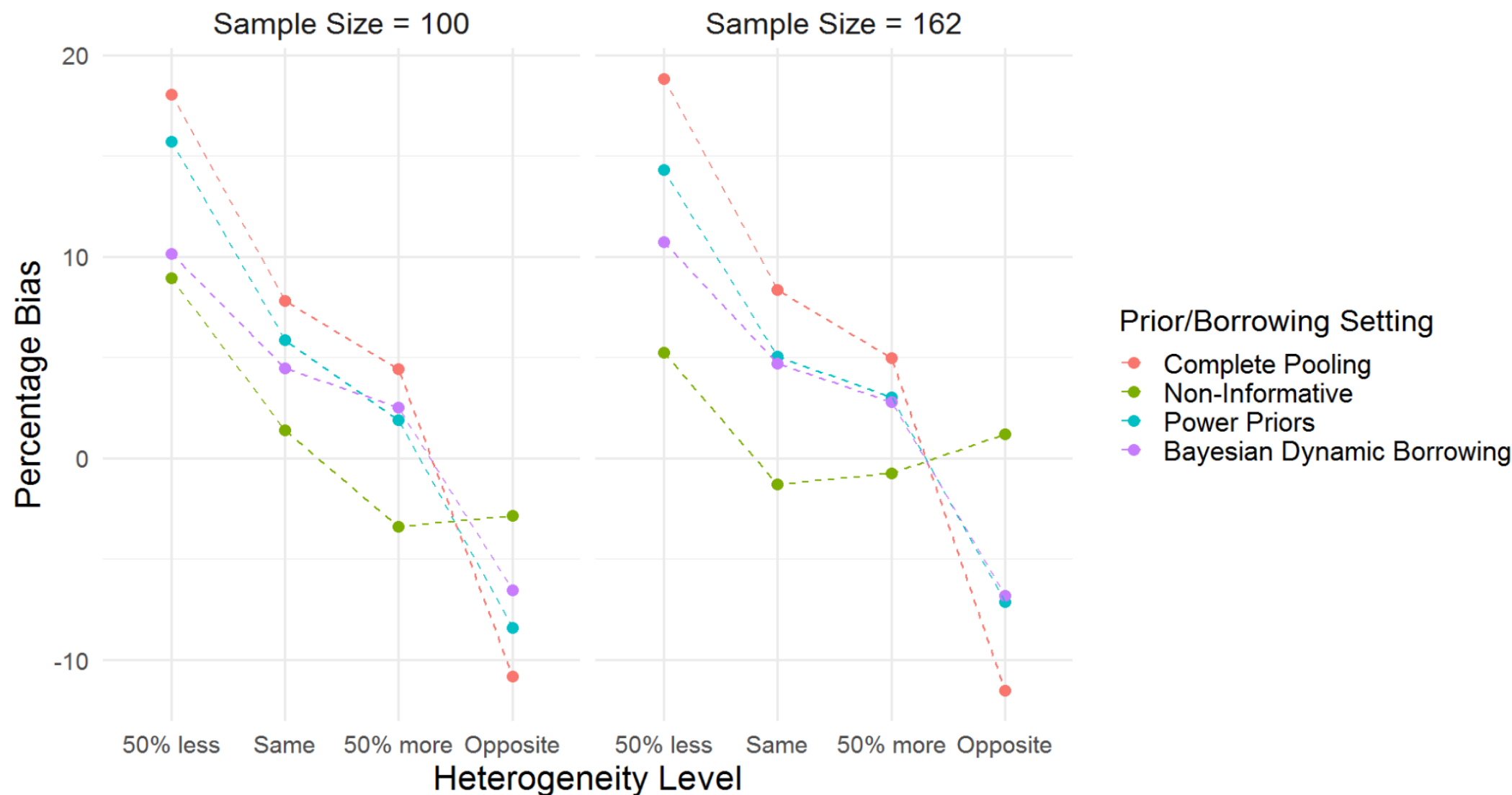
TESS



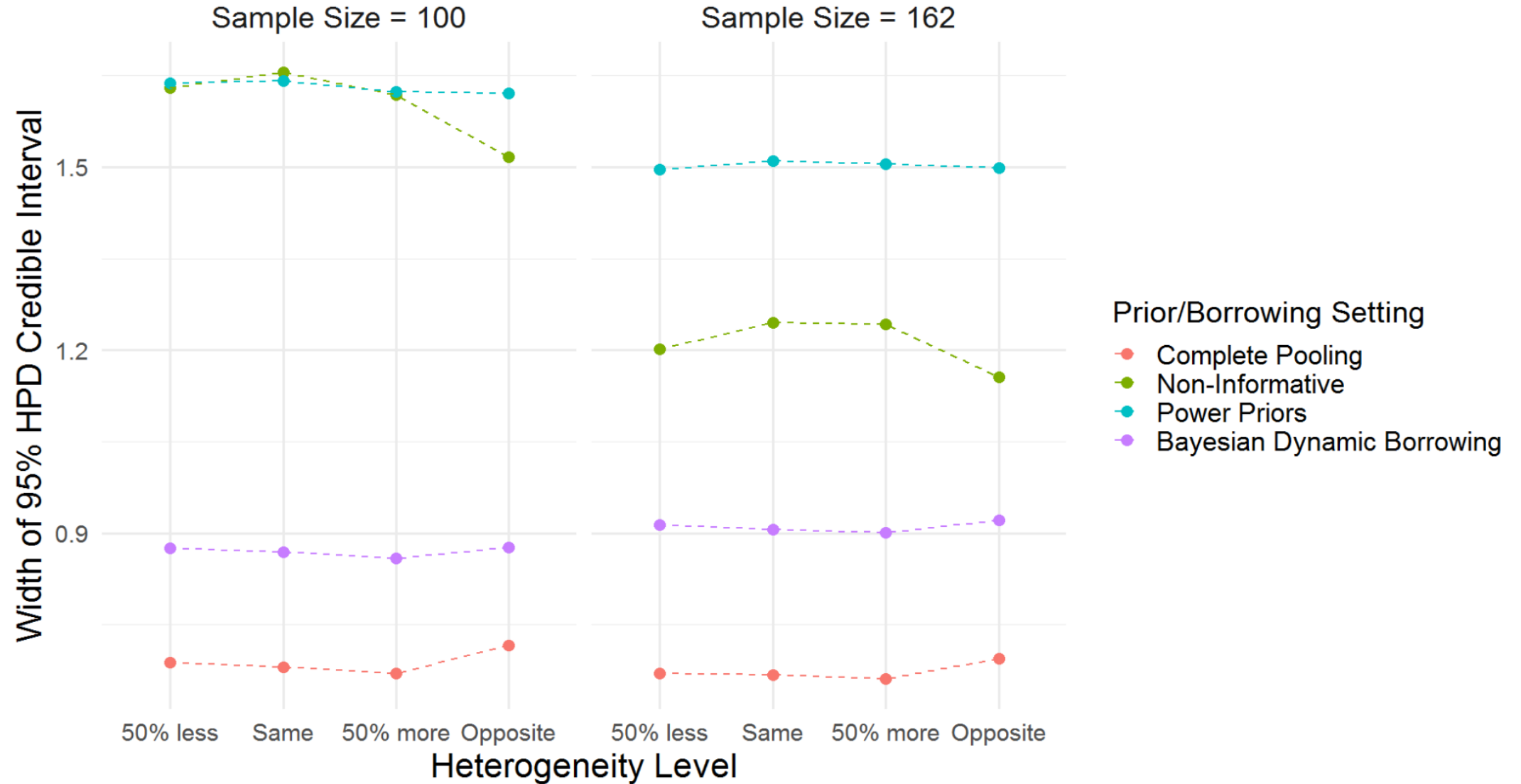
Logarithm of MSE



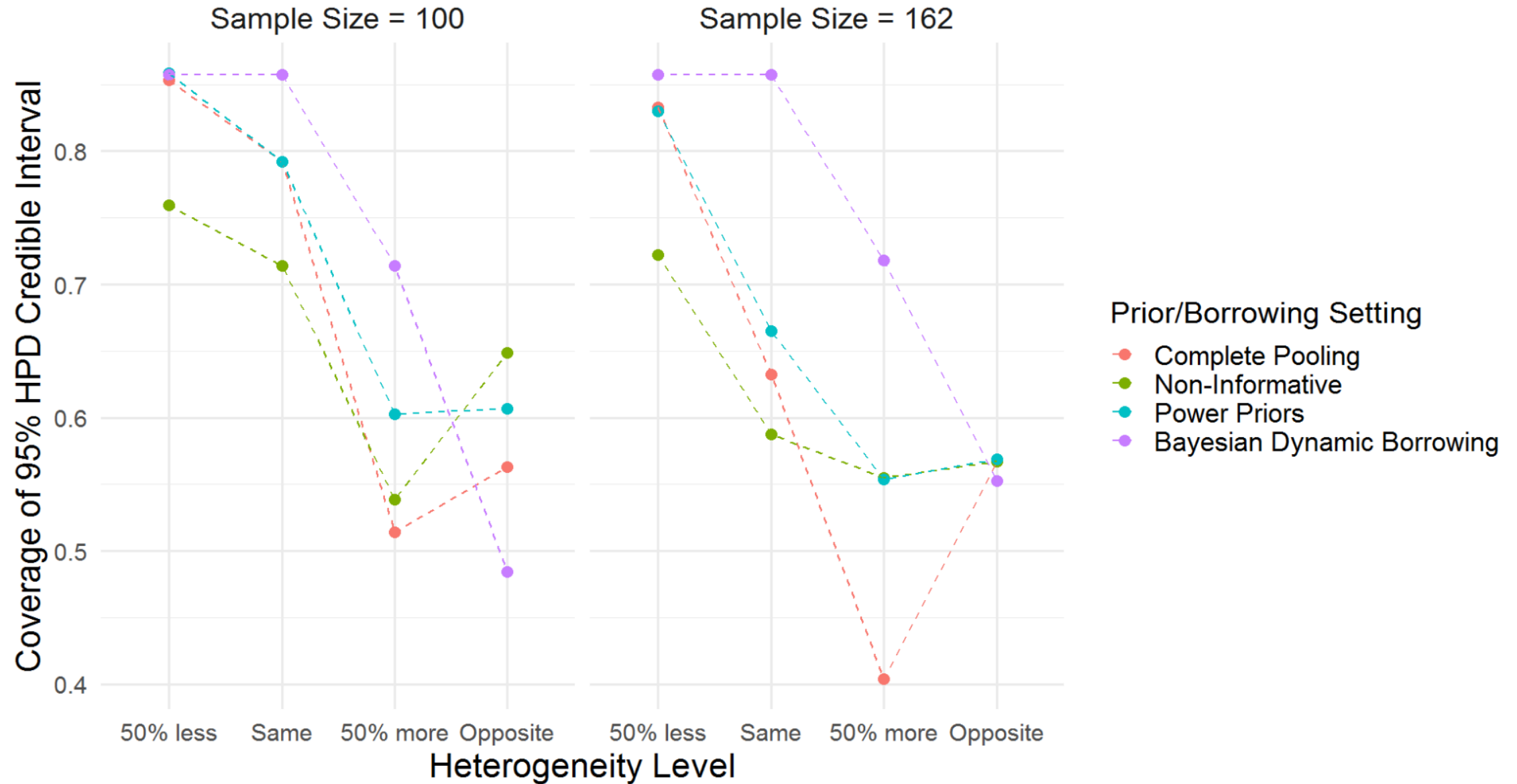
Percentage Bias



Width of 95% HPD CI



Coverage of 95% HPD CI



Conclusions

- BDB performs **worst** in terms of LOOIC across borrowing methods.
- BDB maintained **higher TESS** even in case of smaller sample size.
- BDB achieves **lowest MSE** among other borrowing methods.
- BDB achieves **smaller width for credible intervals** among all priors only next to complete pooling.
- BDB **performs best** in terms of coverage of credible interval. Better than complete pooling which has narrowest width.
- Overall, BDB **performed best** in terms of Width-Coverage tradeoff.

Future Work and Improvements

- Include the standard errors of performance metric values in discussion, to achieve quantification of the uncertainty in our results.
- Include more heterogeneity levels, sample sizes, priors in simulation.
- Studying absolute percentage bias instead of percentage bias.
- Include more prognostic covariates that drives the similarities and dissimilarities among the clinical trails.

Thank You

Any Questions?